

Phys 20AL Week 6 Lab Report: Driven Harmonic Oscillator

Zih-Yu Hsieh

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1 Introduction

When given a damped harmonic oscillator with natural frequency w_0 and damping constant γ , if it's driven by a force of the form $F_0 \cos(wt)$ (where w is the driven frequency), then after a long time (with the system reaches steady state), the position is given with the following function:

$$x \approx \frac{F_0/m}{\sqrt{(w_0^2 - w^2)^2 + 4\gamma^2 w^2}} \cos(wt - \delta), \quad A = \frac{F_0/m}{\sqrt{(w_0^2 - w^2)^2 + 4\gamma^2 w^2}}, \quad \delta = \arctan\left(\frac{2\gamma w}{w_0^2 - w^2}\right) \quad (1)$$

Where A denotes the maximum amplitude, and δ denotes the phase difference. The goal of this experiment is to observe if the experimental data fit with such theoretical prediction, and furthermore extract the damping constant γ based on the collected data of amplitude, frequency, and phase difference.

2 Experimental Setup

The equipments included: A large stand (with multiple clamps provided), a heavy mass of 0.5 kg (used as driven source of the harmonic oscillator), metal wire, multiple strings (total of 12), multiple light masses (total of 11, used as pendulums), scissors, tapes, two meter sticks, and phones.

The setup followed the steps below:

1. Tie the metal wire horizontally on the stand, and ensure it's tight enough.
2. Use one of the strings to tie the heavy mass (0.5 kg) onto the metal wire (near one end of the wire), and tape it to ensure the metal wire would obtain some rotational motion when the heavy mass is swinging.
3. Tie the light masses onto the wire. Ensure that they're spaced (so the pendulum motion would affect each other), while also close enough to the heavy mass (so the driven force caused by the heavy mass wouldn't be dissipated). Also, keep in mind of making each light mass pendulum with a different length, and with at least one having the same length as the heavy mass.
4. Setup the two meter sticks below the pendulums: Fix one directly under the heavy mass, and fix the other meter stick directly under the light mass that's the farthest away from the heavy mass.
6. Fix the phone at the other end of the stand, making sure the camera captures all pendulums, having all the masses are aligned in the center, and also including the meter sticks in the view of the camera.



Figure 1: The Lab Setup Picture

3 Quantity to Measure and Method of Measurement

In this experiment, one recorded a total of three trials. It is aimed to measure two quantities: The amplitudes of all the pendulum, and the phase difference.

For both amplitudes and phase difference, whole experimental process were recorded at stable state using camera. Then, the meter stick will serve as a metric to measure the amplitude of each pendulum; also, the time stamp of the video recording served as a source for measuring the time difference between the moment when each pendulum reaches maximum amplitude (vs. the moment when the driven pendulum reaches its maximum).

Finally, as a preliminary, the used pendulum lengths (cm) are measured using meter sticks, and are given as the following (total of 11 lengths, from the least to the most):

26.3 cm, 30.4 cm, 34.7 cm, 40.1 cm, 42.8 cm, 44.7 cm, 47.4 cm, 49.2 cm, 55.7 cm, 59.8 cm, 64.3 cm.

4 Experimental Procedure

The procedure is as below:

1. Put the whole system at static equilibrium (having none of the pendulums actually moving), and raise the heavy mass to about 10 cm away from equilibrium.
2. Release the heavy mass pendulum, and wait until the system to reach a stable state.
3. When the system reaches stable state, start recording, and let all the pendulums finish at least 5 to 6 cycles.
4. Repeat Step 1 to 3 for a total of three trials.

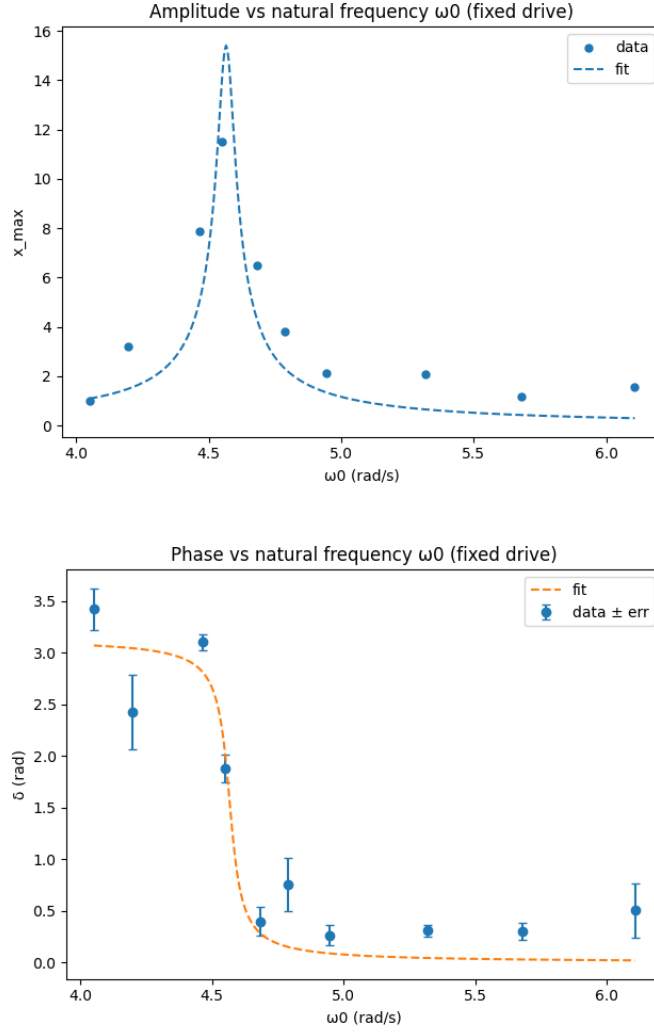
5 Experimental Data and Data Analysis

Here are the experimental data separated by trials. Here, the driven frequency is fixed as $w = 4.669$ rad/s. Also, ω_0 denotes natural frequency (unit: rad/s), $x_{\max}$ (unit: cm) denotes maximum amplitude, and δ denotes average phase difference. Finally, Length here represents each pendulum's length (unit: cm).

5.1 Trial 1

Table 1: Trial 1 Data

Length (cm)	ω_0	$x_{\max}$	δ	$x_{\max, err}$	δ_{err}
26.300	6.107	1.537	0.504	0.237	0.264
30.400	5.681	1.174	0.298	0.308	0.084
34.700	5.317	2.082	0.306	0.039	0.060
40.100	4.946	2.102	0.260	1.101	0.098
42.800	4.788	3.786	0.754	0.188	0.254
44.700	4.685	6.486	0.398	0.572	0.144
47.400	4.549	11.520	1.877	1.949	0.132
49.200	4.465	7.851	3.103	0.614	0.080
55.700	4.197	3.196	2.424	0.789	0.359
59.800	4.050	1.000	3.422	0.423	0.203
64.300	3.906	1.500	3.301	0.294	0.681



As a result, the calculated damping constant (using the provided program) is $\gamma_1 = 0.0351$ kg/s.

This result (including the data and graphs) are mostly as expected, since the original driven frequency is given as 4.669 rad/s, so the expected maximum of amplitude should be around this driven frequency. For the amplitude graph one can see the peak is around when w_0 is 4.6 – 4.7 rad/s, which is as expected. Also, it demonstrated a rapid increase when w_0 is smaller than driven frequency, while rapidly decreasing for w_0 being greater than driven frequency, which is also as predicted by the formula.

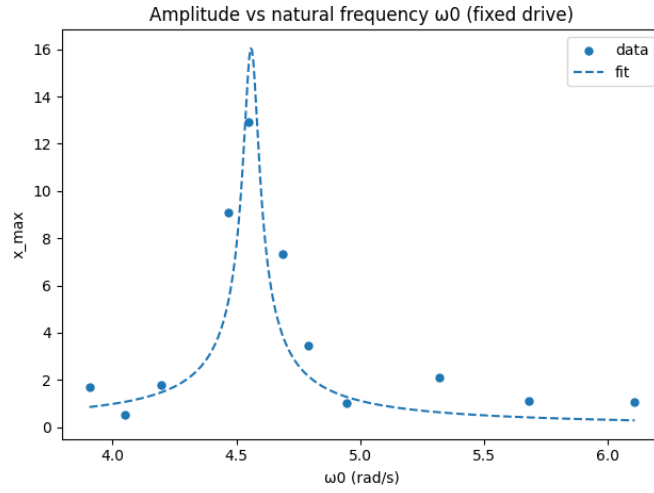
Similarly, when looking at the phase graph, one can see that there is also a similar curve (nearly flat at the start, a rapid drop around driven frequency 4.669 rad/s, then goes back to a flat curve). Notice that the two flat parts are about 3 apart, which is pretty similar to the predicted gap of $\pi \approx 3.14$; also, when w_0 is around 4.6 – 4.7 rad/s, one can see the δ value is about $\frac{\pi}{2}$, this is also as predicted that around the driven frequency, the pendulum has a $\frac{\pi}{2}$ phase difference from the driven source.

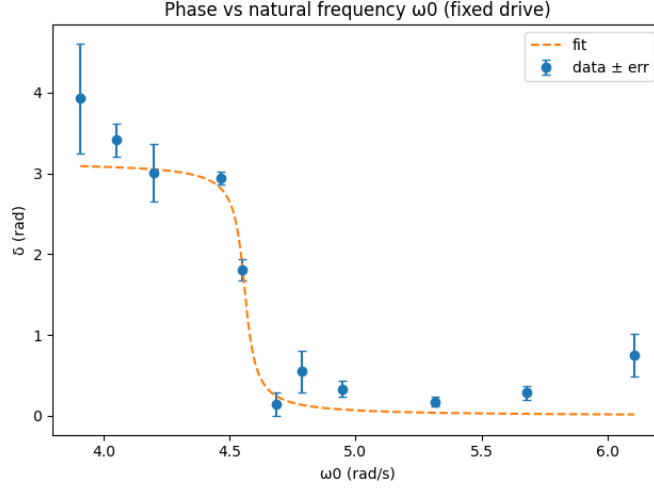
However, here because of the way of calculating phase difference, the graph instead of starting at 0 near $w_0 = 0$ (as the predicted graph), it starts at ≈ 3 near $w_0 = 0$. But since the pattern is mostly following the expectation, so one still has a high confidence of saying that after a shift, the data mostly fits with the predicted graph.

5.2 Trial 2

Table 2: Trial 2 Data

Length (cm)	ω_0	$x_{\max}$	δ	$x_{\max, \text{err}}$	δ_{err}
26.3	6.107	1.061	0.748	0.237	0.264
30.4	5.681	1.122	0.284	0.308	0.084
34.7	5.317	2.082	0.173	0.039	0.060
40.1	4.946	1.012	0.334	1.101	0.098
42.8	4.788	3.458	0.551	0.188	0.254
44.7	4.685	7.326	0.141	0.572	0.144
47.4	4.549	12.920	1.808	1.949	0.132
49.2	4.465	9.088	2.947	0.614	0.080
55.7	4.197	1.798	3.011	0.789	0.359
59.8	4.050	0.500	3.412	0.423	0.203
64.3	3.906	1.700	3.926	0.294	0.681





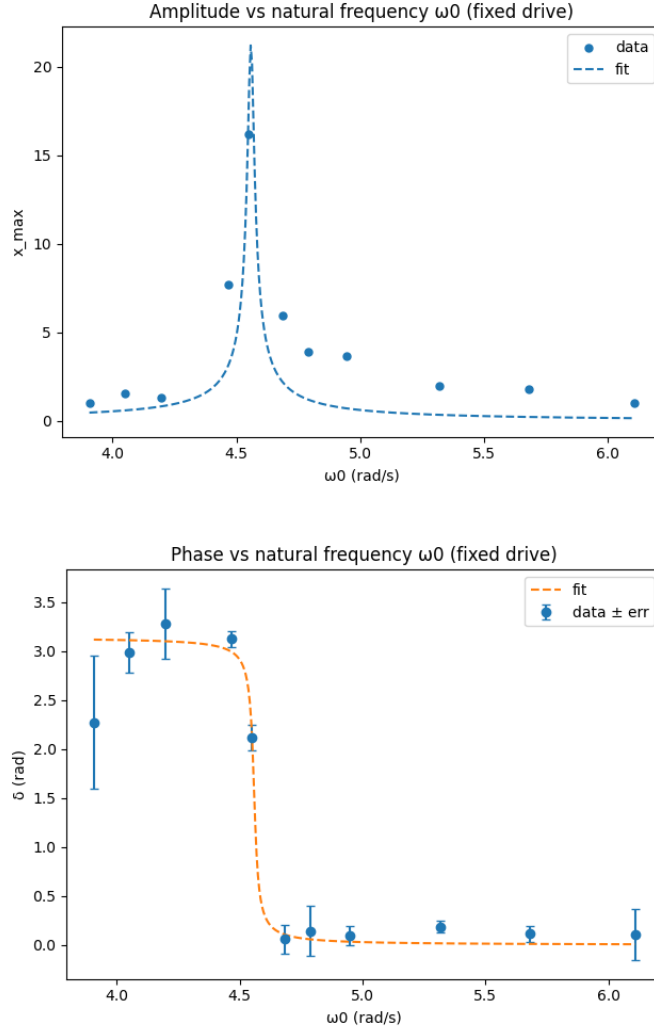
As a result, the calculated damping constant is $\gamma_2 = 0.0328$ kg/s.

Here, similar to Trial 1, one can see that the graphs of amplitude and phase both have a similar shape to the predicted equation; furthermore, one can also see that when w_0 is about 4.6–4.7 rad/s (when pretty close to the driven frequency), the graph of amplitude is at its maximum, while the graph of phase difference is at about 1.5 (which is close to $\frac{\pi}{2}$), showing that the resonance frequency is also close to the driven frequency $w = 4.669$ rad/s, and it has a phase difference of approximately $\frac{\pi}{2}$, which is also fitting with the prediction.

5.3 Trial 3

Table 3: Experimental data for ω_0 , $x_{\max}$, and δ as a function of length.

Length (cm)	ω_0	$x_{\max}$	δ	$x_{\max, \text{err}}$	δ_{err}
26.3	6.107	1.012	0.107	0.237	0.264
30.4	5.681	1.800	0.114	0.308	0.084
34.7	5.317	2.000	0.186	0.039	0.060
40.1	4.946	3.693	0.099	1.101	0.098
42.8	4.788	3.902	0.144	0.188	0.254
44.7	4.685	5.935	0.059	0.572	0.144
47.4	4.549	16.172	2.115	1.949	0.132
49.2	4.465	7.727	3.126	0.614	0.080
55.7	4.197	1.340	3.284	0.789	0.359
59.8	4.050	1.536	2.987	0.423	0.203
64.3	3.906	1.000	2.275	0.294	0.681



As a result, the calculated damping constant is $\gamma_3 = 0.0138$.

Here, similar to the previous Trials, both the amplitude and phase diagrams are similar to the predicted equation; and, when w_0 is about $4.6 - 4.7$ rad/s (when close th driven frequency), not only amplitude is at its maximum, and the phase difference is also about 1.5 (which is approximately $\frac{\pi}{2}$), this again shows that the data is fitting the prediction in general.

6 Error Analysis

Besides human mistakes and the imprecision in measurement (due to human eyes' limitation), there are some potential sources of error:

- First, the pendulum is not a simple harmonic oscillator. For the possibility of measuring the amplitude, the initial amplitude of the driving source (the pendulum with heavy weight) needs to be high enough in case to be observable. As a consequence, when different pendulums are at different maximum amplitude, the nontrivial effect of period (which is dependent to maximum amplitude) could occur.
- Second, the driven force itself is also damped, so it cannot be perfectly modeled as $F_0 \cos(\omega t)$ (since the amplitude would also decrease over time). Such effect could also affect the collected data.
- Finally, the frame of the camera is also essential: For the camera used in this experiment, it's only up to 10 ms (for ease of data extraction). If the frame is higher it could provide a more accurate data.

As of improvement, if time permitted using a camera with higher frame per second could increase the accuracy of the data; also, if the driving source has an extra kicking mechanism (like the pendulum clock), this could also increase the precision of the experiment (since it could make the pendulum fitting more to a simple harmonic oscillator), but this again depends on the availability (since such mechanism is a somewhat nontrivial one to make in a 3-hour lab).

7 Conclusion

In the experiment, for all three trials one can observe that the experimental data in general have their pattern fit with the formula prediction, hence there is a decent confidence one can say that the formula is a good approximation of the scenario happened in the experiment.

For the calculated damping constant (which is extracted in each trial, instead of separated for amplitude and phase) is given by $\gamma_1 = 0.0351$, $\gamma_2 = 0.0328$, and $\gamma_3 = 0.0138$ kg/s, which actually varies a bit. The calculated average damping constant is $\gamma_{\text{ave}} \approx 0.0272$ kg/s, with a Standard Deviation of $\sigma \approx 0.0117$ kg/s. So, because σ is relatively large in comparison to the scale of the data (which is about 0.01 kg/s as standard scale), this average has a relatively low validity of representing the actual damping constant of the system.

Finally, some sources of error include: The pendulum is not a Simple Harmonic Oscillator (so the maximum amplitude in fact affects the period), the Drivign Source is also a damped oscillator itself (so the driving force also has decreasing amplitude), and possibly the lower frame of the camera used in this experiment.